

The feasibility of a whole-cell model of the human platelet

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Quick recap

What I evaluated

- Aiming to build a computer model of the platelet.
- Has been done before in prokaryotes and yeast but not in other eukaryotic cells.
- Lots of modelling done in platelets but not a whole-cell attempt.
- Lots of options e.g., COPASI, V-Cell and others.
- Landed on wcEcoli project from Covert lab — capable of scaling to whole cell, code quality excellent, something I could work with and expand.

What I did

- Forked wcEcoli (i.e., took my own working copy).
- Pruned all *E. coli* biology.
- Used AI to build tests to verify I didn't remove too much.
- Rebuilt cell contents and processes from the platelet literature.
- Kept the simulation engine, state partitioning, listener / analysis framework.
- ~5,000 lines of working code I didn't have to write.
- Emailed Prof. Covert — “sounds fun, good luck”.

Method to build a calibrated model

For each mechanism, a five-step process:

1. **Anchor paper.** E.g., Dolan & Diamond 2014 supplied the validation experiment (Fig. 4 Ca^{2+} transients $\pm$ extracellular Ca^{2+}) and resting-state targets (100 nM cyt, 250 μM DTS).
2. **Literature review for kinetics.** Primary sources: deYoung–Keizer 1992 for IP_3R , Caride 2007 for PMCA, Hoover & Lewis 2011 for SOCE, Mazet 2020 for the PI cycle...
3. **Species enumeration.** Every Ca^{2+} -binding or -gating protein state, with a compartment tag (DTS, Cyt etc.). ~50 species for the calcium pathway.
4. **Copy numbers** from Burkhart 2012 platelet proteomics.
5. **Rate constants.** Primary-source values, units normalised, every value carrying a citation comment in code.

Then: does the resting state hold? Do the timescales line up? Do the unit tests still pass?

Build code from source data — start with the data

Start with the published data (usually PDF, sometimes CSV).

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TABLE 1 Reaction equations, rate laws, and kinetic constants

	Reaction / Quantity	Mechanism	Rate Law / Rule	Parameter Values	Ref.
SERCA	SERCA shuttling	$SERCA_{E2} \leftrightarrow SERCA_{E1}$	$k_1 \cdot [SERCA_{E2}] - k_{-1} \cdot [SERCA_{E1}]$	$k_1 = 600 \text{ s}^{-1}, k_{-1} = 600 \text{ s}^{-1}$	(11)
	Ca^{2+}_{cyt} binding SERCA	$SERCA_{E1} + 2 \text{ } Ca^{2+}_{\text{cyt}} \leftrightarrow SERCA_{E1} \cdot Ca^{2+}$	$k_1 \cdot [SERCA_{E1}] \cdot [Ca^{2+}_{\text{cyt}}]^2 - k_{-1} \cdot [SERCA_{E1} \cdot Ca^{2+}]$	$k_1 = 1 \times 10^{15} \text{ M}^{-2} \text{ s}^{-1}, k_{-1} = 10 \text{ s}^{-1}$	(11)
	Phosphorylation of SERCA	$SERCA_{E1} \cdot Ca^{2+} \leftrightarrow SERCA_{E1}P \cdot Ca^{2+}$	$k_1 \cdot [SERCA_{E1} \cdot Ca^{2+}] - k_{-1} \cdot [SERCA_{E1}P \cdot Ca^{2+}]$	$k_1 = 700 \text{ s}^{-1}, k_{-1} = 5 \text{ s}^{-1}$	(11)
	Ca^{2+} transport across IM	$SERCA_{E1}P \cdot Ca^{2+} \leftrightarrow SERCA_{E2}P \cdot Ca^{2+}$	$k_1 \cdot [SERCA_{E1}P \cdot Ca^{2+}] - k_{-1} \cdot [SERCA_{E2}P \cdot Ca^{2+}]$	$k_1 = 600 \text{ s}^{-1}, k_{-1} = 50 \text{ s}^{-1}$	(11)
	Ca^{2+}_{dts} release into DTS	$SERCA_{E2}P \cdot Ca^{2+} \leftrightarrow SERCA_{E2}P + 2 \text{ } Ca^{2+}_{\text{dts}}$	$k_1 \cdot [SERCA_{E2}P \cdot Ca^{2+}] - k_{-1} \cdot [SERCA_{E2}P] \cdot [Ca^{2+}_{\text{dts}}]^2$	$k_1 = 1000 \text{ s}^{-1}, k_{-1} = 4 \times 10^9 \text{ M}^{-2} \text{ s}^{-1}$	(11)
	SERCA dephosphorylation	$SERCA_{E2}P \leftrightarrow SERCA_{E2}$	$k_1 \cdot [SERCA_{E2}P] - k_{-1} \cdot [SERCA_{E2}]$	$k_1 = 500 \text{ s}^{-1}, k_{-1} = 1 \text{ s}^{-1}$	(11)
IP ₃ R	IP ₃ R inhibition	$IP_3R_n + Ca^{2+}_{\text{cyt}} \leftrightarrow IP_3R_{n1}$	$[IP_3R_n] \cdot ((k_1 \cdot L_1 + l_2) \cdot [Ca^{2+}_{\text{cyt}}] / (L_1 + [Ca^{2+}_{\text{cyt}}] \cdot (1 + L_1 / L_3))) - [IP_3R_{n1}] \cdot (k_{-1} + l_{-2})$	$k_1 = 0.64 \text{ s}^{-1} \cdot \mu\text{M}^{-1}, L_1 = 0.12 \mu\text{M}, l_2 = 1.7 \text{ s}^{-1}$ $L_3 = 0.025 \mu\text{M}, k_{-1} = 0.04 \text{ s}^{-1}, l_{-2} = 0.8 \text{ s}^{-1}$	(12)
	IP ₃ R binding IP ₃	$IP_3R_n + IP_3 \leftrightarrow IP_3R_o$	$[IP_3R_n] \cdot [IP_3] \cdot ((k_2 \cdot L_3 + l_4 \cdot [Ca^{2+}_{\text{cyt}}]) / (L_3 + [Ca^{2+}_{\text{cyt}}] \cdot (1 + L_3 / L_1))) - [IP_3R_o] \cdot ((k_{-2} + l_{-4} \cdot [Ca^{2+}_{\text{cyt}}]) / (1 + [Ca^{2+}_{\text{cyt}}] / L_5))$	$k_2 = 37.4 \text{ s}^{-1} \cdot \mu\text{M}^{-1}, l_4 = 1.7 \text{ s}^{-1} \cdot \mu\text{M}^{-1}$ $k_{-2} = 1.4 \text{ s}^{-1}, l_{-4} = 2.5 \mu\text{M}^{-1} \cdot \text{s}^{-1}, L_5 = 54.7 \mu\text{M}$	(12)
	IP ₃ R activation	$IP_3R_o + Ca^{2+}_{\text{cyt}} \leftrightarrow IP_3R_a$	$[IP_3R_o] \cdot ((k_4 \cdot L_5 + l_6) \cdot [Ca^{2+}_{\text{cyt}}] / (L_5 + [Ca^{2+}_{\text{cyt}}])) - [IP_3R_a] \cdot (L_1 \cdot (k_{-4} + l_{-6}) / (L_1 + [Ca^{2+}_{\text{cyt}}]))$	$k_4 = 4 \text{ s}^{-1} \cdot \mu\text{M}^{-1}, l_6 = 4707 \text{ s}^{-1}$ $k_{-4} = 0.54 \text{ s}^{-1} \cdot \mu\text{M}^{-1}, l_{-6} = 11.4 \text{ s}^{-1}$	(12)
	IP ₃ R inhibition	$IP_3R_a + Ca^{2+}_{\text{cyt}} \leftrightarrow IP_3R_{i2}$	$[IP_3R_a] \cdot (k_1 \cdot L_1 + l_2) \cdot [Ca^{2+}_{\text{cyt}}] / (L_1 + [Ca^{2+}_{\text{cyt}}]) - [IP_3R_{i2}] \cdot (k_{-1} + l_{-2})$		(12)
	IP ₃ R closing	$IP_3R_o \leftrightarrow IP_3R_s$	$[IP_3R_o] \cdot (k_3 \cdot L_5 / (L_5 + [Ca^{2+}_{\text{cyt}}])) - [IP_3R_s] \cdot k_{-3}$	$k_3 = 11 \text{ s}^{-1} \cdot \mu\text{M}^{-1}, k_{-3} = 29.8 \text{ s}^{-1}$	(12)
	Channel open probability (P_{oIP3R})		$(0.9 \cdot IP_3R_a / IP_3R_{\text{total}} + 0.1 \cdot pIP_3R_o / IP_3R_{\text{total}})^4$		(12)
	IM potential (ψ_{IM})		$RT / zF \cdot \ln (Ca^{2+}_{\text{cyt}} / Ca^{2+}_{\text{dts}})$		
	Ca^{2+} release from DTS via IP ₃ R	$Ca^{2+}_{\text{dts}} \leftrightarrow Ca^{2+}_{\text{cyt}}$	$N_{\text{IP3R}} \cdot P_{\text{oIP3R}} \cdot Y_{\text{IP3R}} \cdot e \cdot (V_{\text{IM}} - \psi_{\text{IM}})$	$Y_{\text{IP3R}} = 10 \text{ pS}$	(30)
	Ca^{2+}_{dts} binding STIM1	$STIM1 + Ca^{2+}_{\text{dts}} \leftrightarrow STIM1 \cdot Ca^{2+}_{\text{dts}}$	$k_1 \cdot [STIM1] \cdot [Ca^{2+}_{\text{dts}}] - k_{-1} \cdot [STIM1 \cdot Ca^{2+}_{\text{dts}}]$	$k_{-1} / k_1 = 200 \mu\text{M}$	(1)
	STIM1 dimerization	$2 \text{ } STIM1 \leftrightarrow STIM1_2$	$k_1 \cdot [STIM1]^2 - k_{-1} \cdot [STIM1_2]$	$k_1 = 9 \times 10^8 \text{ s}^{-1} \cdot \text{M}^{-1}, k_{-1} = 3.5 \text{ s}^{-1}$	This study
STIM	Puncta formation factor (θ_p)		$\alpha \cdot [Ca^{2+}_{\text{cyt}}]^n / ([Ca^{2+}_{\text{cyt}}]^n + K_M^n) + 0.01$	$K_M = 100 - 300 \text{ nM}, n = 0.5 - 3, \alpha = 0.2$	This study
	STIM1 ₂ puncta entry ($(STIM1_2)_p$)	$STIM1_2 \leftrightarrow (STIM1_2)_p$	$[STIM1_2] \cdot \theta_p$		
	STIM in puncta binding Orai1	$(STIM1_2)_p + \text{Orai1} \leftrightarrow (STIM1_2)_p \cdot \text{Orai1}$	see equations below		

Just a small excerpt.

Build code from source data — AI extraction

Ask AI to extract — get it to tell you what it's doing to sanity check (but it is very good at this).

E.g. I asked: “Please read Dolan and Diamond 2014 and tell me what the first 5 lines of table 1 show.”

Simply not feasible to do this by hand at large scale; AI could process 100s of papers in a day. The only challenge is verification, but this is also amenable to AI assistance.

Read 1 file, listed 1 directory (ctrl+o to expand)

• Table 1 is titled “Reaction equations, rate laws, and kinetic constants” (p. 2852). The first 5 rows are all in the SERCA module and reference (11) – the Dode 2002 SERCA paper. They lay out the E1/E2 cycle one step per row.

#	Reaction / Quantity	Mechanism	Rate Law	Parameter values
1	SERCA shuttling	$SERCA_E2 \rightarrow SERCA_E1$	$k_1 \cdot [SERCA_E2] - k_{-1} \cdot [SERCA_E1]$	$k_1 = 600 \text{ s}^{-1}$, $k_{-1} = 600 \text{ s}^{-1}$
2	Ca^{2+} binding SERCA	$SERCA_E1 + 2 Ca^{2+}_{cyt} \rightarrow SERCA_E1 \cdot Ca$	$k_2 \cdot [SERCA_E1] \cdot [Ca^{2+}_{cyt}]^2 - k_{-2} \cdot [SERCA_E1 \cdot Ca]$	$k_2 = 1 \times 10^6 \text{ M}^{-1} \text{ s}^{-1}$, $k_{-2} = 10 \text{ s}^{-1}$
3	Phosphorylation of SERCA	$SERCA_E1 \cdot Ca^{2+} \rightarrow SERCA_E1P \cdot Ca$	$k_3 \cdot [SERCA_E1 \cdot Ca^{2+}]$	$k_3 = 700 \text{ s}^{-1}$
4	Ca^{2+} transport across IM	$SERCA_E1P \cdot Ca^{2+} \rightarrow SERCA_E2P \cdot Ca$	$k_4 \cdot [SERCA_E1P \cdot Ca^{2+}]$	$k_4 = 5 \text{ s}^{-1}$
5	Ca^{2+} release into DTS	$SERCA_E2P \cdot Ca^{2+} \rightarrow SERCA_E2P + 2 Ca^{2+}_{dts}$	$k_5 \cdot [SERCA_E2P \cdot Ca^{2+}] - k_{-5} \cdot [SERCA_E2P] \cdot [Ca^{2+}_{dts}]^2$	$k_5 = 1000 \text{ s}^{-1}$, $k_{-5} = 4 \times 10^5 \text{ M}^{-1} \text{ s}^{-1}$

So: the full forward E1-E1-Ca-E1P-Ca-E2P-Ca-E2P-E2 cycle in five rows, with the loop-closing E2-E1 reshuffle as row 1. These map onto your K_SERCA constants in calcium_signalling.py and the six SERCA sub-state species in internal_state.py (SERCA_E1, SERCA_E2, SERCA_E1Ca, SERCA_E1Pca, SERCA_E2Pca, SERCA_E2P).

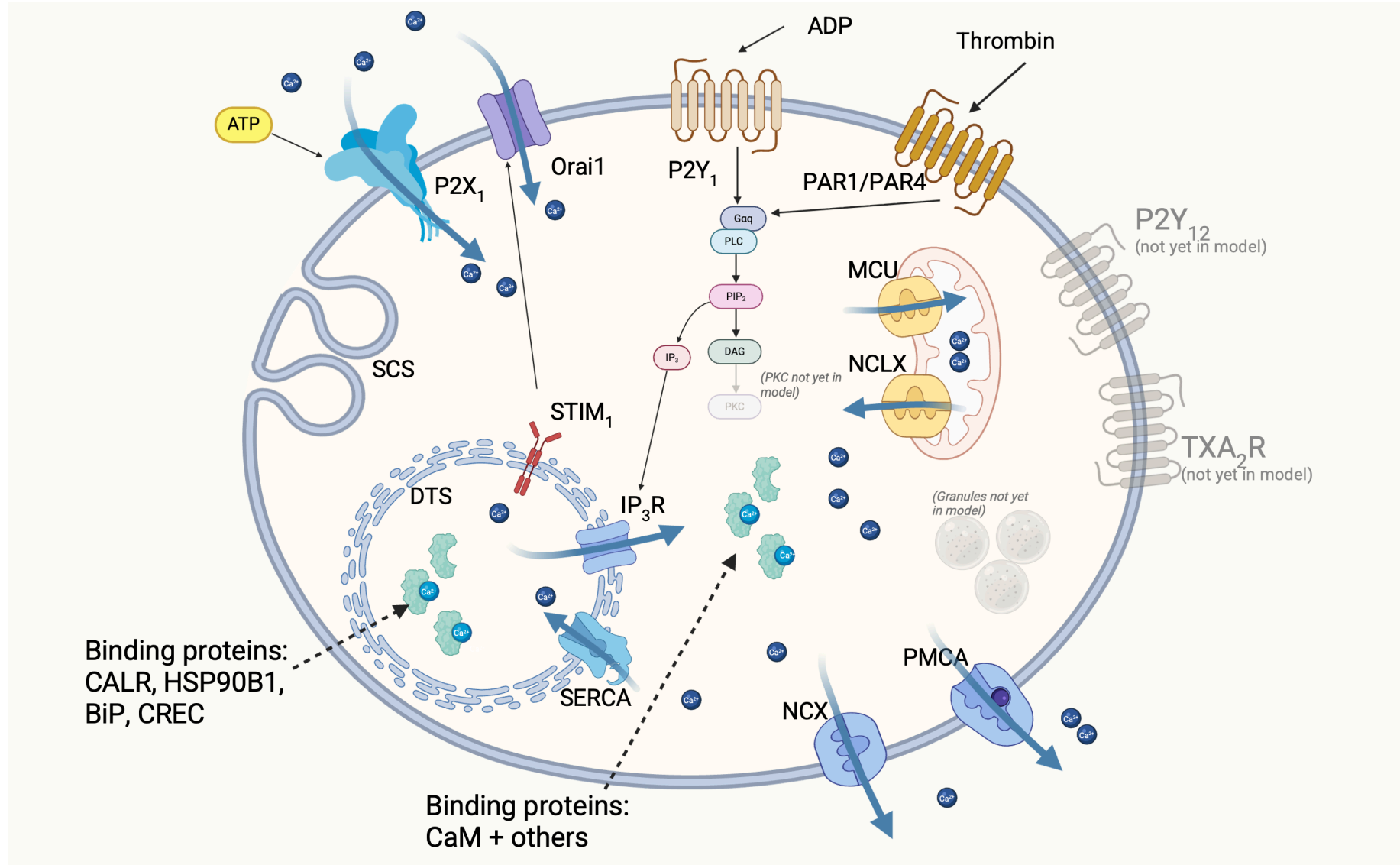
Build code from source data — AI writes the code

Then get AI to build the code, using the *wcEcoli* code as a template.

The code works well, the architecture is not perfect — needs some work to make it more usable for future expansion — but it's an excellent start.

```
# IP3R, SERCA, PMCA, STIM1, and CaM are split into their kinetic sub-states so
# the CalciumDynamics ODE solver can integrate the 6-state Sneyd IP3R, the SERCA
# E1/E2 cycle, the 5-state Caride 2007 CaM-coupled PMCA scheme, the STIM1
# DTS-bound / free / dimer sensor cycle, and the 3-state CaM Ca2+-binding ladder.
# Sub-state initial counts are the Dolan & Diamond 2014 Table S1 representative
# configuration; protein totals match the per-protein totals in that table
# (1,328 IP3R; 11,892 SERCA; 769 PMCA; 4,265 STIM1; 1,447 Orail; 20,481 CaM).
_MOLECULES = [
  # id          mass_fg      initial_count  molecule_class
  # — metabolites (concentrations derived from Purvis 2008 / Dolan 2014 / Sveshnikova 2025) —
  ('CA2_CYT[c]', 6.660e-8, 361, 'metabolite'), # 100 nM × 6 fL
  ('CA2_DTS[ds]', 6.660e-8, 38842, 'metabolite'), # 250 μM × 4.3% × 6 fL
  ('ATP[c]', 8.424e-7, 10_839_600, 'metabolite'), # 3 mM × 6 fL (Holmsen 1979/1981)
  ('ADP[c]', 7.096e-7, 1_083_960, 'metabolite'), # 0.3 mM × 6 fL (ATP:ADP = 10:1)
  ('PI[c]', 1.580e-7, 361_320, 'metabolite'), # 100 μM × 6 fL inorganic phosphate
  ('5HT[dg]', 2.927e-7, 3_500_000, 'metabolite'), # serotonin; dense granule
  ('ADP[dg]', 7.096e-7, 400_000, 'metabolite'), # ADP; dense granule
  ('IP3[c]', 6.977e-7, 181, 'metabolite'), # 50 nM × 6 fL
  # — proteins (copy numbers from Burkhardt 2012 unless noted) —
  ('GP1BA[c]', 1.378e-4, 25_000, 'protein'), # GpIba; surface receptor
  ('ITGA2B[c]', 2.149e-4, 80_000, 'protein'), # αIIb integrin
  ('ACTB[c]', 6.933e-5, 2_000_000, 'protein'), # β-actin
  ('FGA[ag]', 5.647e-4, 30_000, 'protein'), # fibrinogen hexamer; alpha granule
  ('SELP[ag]', 1.493e-4, 30_000, 'protein'), # P-selectin; alpha granule
```

What did I build? (so far)



Spotting errors in the source

The Purvis 2008 k_3 story.

- Asked AI to check values it had extracted.
- AI spotted that a value quoted in Purvis & Diamond 2008 (a well-known platelet modelling paper) was wrong — units were wrong and value was 100× too large.
- AI found the original study via the references and I verified for myself; AI was correct, there was a typo.
- Very strong use-case for AI — tracking values and assertions across papers. Also very useful for helping to review data: I got the AI to print values it had found, with reference to the primary source and the code, so I could easily review.

Validation

Criteria

Compared output to existing models.

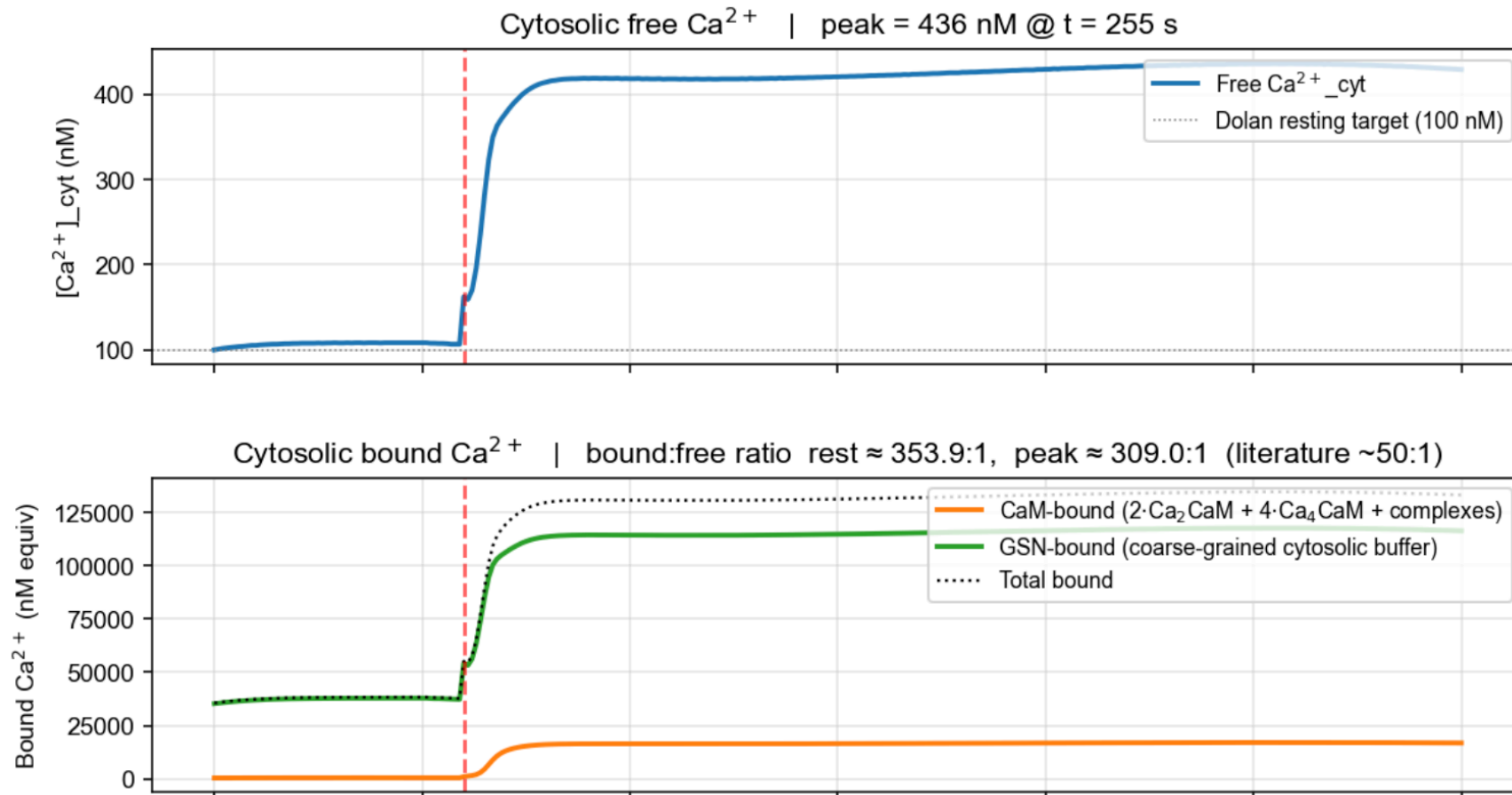
Success criteria:

- Resting cytosolic Ca^{2+} within 100 ± 10 nM band
- Resting DTS Ca^{2+} in the literature range
- IP_3 -stimulated transient peak height in band
- Transient duration matches shape of previous model(s)

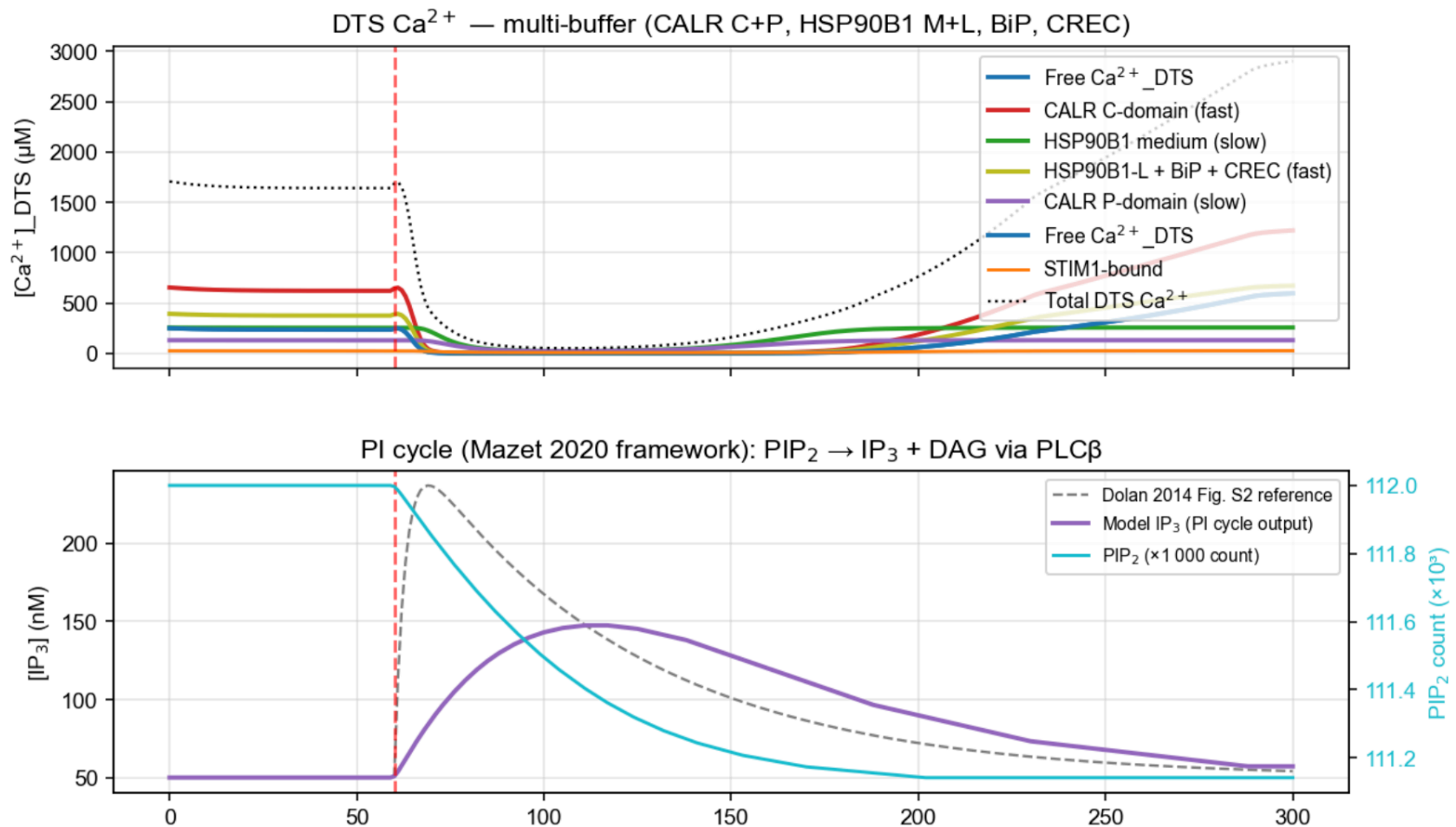
Model state at 14 May

- Resting cyt Ca^{2+} : 104 nM ✓
- Resting DTS Ca^{2+} : 235 μM ✓
- Resting IP_3 : 50 nM ✓
- Resting G α q-active: 100 of 5000 ✓
- 21 / 21 unit tests pass
- Driven by physiological agonists — 1 nM thrombin, 10 μM ADP. No hand-fitted IP_3 forcing.

Model stimulated with ADP / thrombin — cytosolic Ca^{2+}



Model stimulated with ADP / thrombin — DTS Ca^{2+}



Where it goes from here

Near-term biology

- Granule release — the model currently stops at the Ca^{2+} peak.
- Implement P2Y_{12} , TXA2R receptors and more, depending on time.
- Run experiments e.g. when $\text{P2Y}_{12} / \text{G}_i$ are wired up, add clopidogrel as an *in silico* drug experiment.

Wider impact

- Platelet–CTC interactions. A calibrated platelet model could in principle be coupled to a tumour-cell model.
- Methodology generalises to other single-cell calibration problems — if pathways are well-quantified, no reason not to try modelling them in a cancer cell.
- Multiple cell–cell interactions — thrombus formation?

Blue sky goal

- I'd like to get to a point where a biologist with limited coding experience could build a model with AI help. Very long shot: a system where a user drops primary data and text instructions into an app and the model builds itself.